

RAUNAK DEY



IIT Madras
B.S. Degree

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SUMMARY

Data Science and Machine Learning student with experience in AI systems, information retrieval, quantitative research, and full-stack engineering. Builds and evaluates data-driven systems using machine learning, embeddings, retrieval pipelines, and statistical experimentation.

EDUCATION

Indian Institute of Technology Madras
BS in Data Science and Applications | CGPA: 7.21/10

Chennai, Tamil Nadu
September 2022 – August 2027

SKILLS

Languages: Python, SQL, TypeScript, JavaScript
Frameworks: FastAPI, Next.js, React, Flask

AI/ML: PyTorch, Scikit-learn, Sentence Transformers
Data & Infrastructure: PostgreSQL, pgvector, Redis, Docker

EXPERIENCE

Quantitative Research Consultant
WorldQuant

April 2024 – March 2025
Remote

- Developed and evaluated 15 quantitative alphas across price-volume, news, options, and social-media datasets, using 25 unique operators and 28 financial data fields
- Completed 4 alpha pyramids spanning News, Options, Price-Volume, and Social Media datasets
- Achieved combined alpha performance of 0.82 and Power Pool performance of 1.12 through iterative signal research and evaluation

PROJECTS

Mind Palace | *Python, FastAPI, PostgreSQL, pgvector, MCP*

August 2026 – Present

- Built an open-source AI memory infrastructure that transforms knowledge corpora into attributable, model-ready context through incremental ingestion, embeddings, hybrid retrieval, and token-budgeted context packing
- Implemented vector, keyword, and Reciprocal Rank Fusion (RRF) retrieval using pgvector and pg_trgm, achieving 87% Recall@5 at ~51ms latency on a 202-document benchmark
- Designed an evaluation and quality-gating pipeline over 98 queries across 14 categories using Recall@K, MRR, nDCG, and paired bootstrap confidence intervals to detect retrieval regressions

Finalysis 2.0 | *Next.js 16, TypeScript, Redis, Tailwind CSS*

May 2025 – June 2026

- Built an investor-facing stock research platform covering 2,300+ NSE equities with fundamental metrics, price momentum, news sentiment, and explainable recommendations
- Integrated Yahoo Finance, Screener.in, Google News RSS, Redis caching, and daily market snapshots into a resilient multi-source data pipeline
- Engineered fault-tolerant serverless APIs with rate limiting, circuit breakers, cache fallbacks, and automated price-refresh workflows, deployed at fin-alysis.vercel.app

CV-Embed | *React 19, TypeScript, Vite, @react-pdf/renderer*

February 2026 – July 2026

- Built an ATS-friendly resume builder with real-time preview, structured editing, browser persistence, and configurable templates, deployed at cv-embed.vercel.app
- Engineered PDF, DOCX, and normalized JSON export pipelines with Zod-based validation and automated resume quality scoring
- Developed an embeddable SDK v2 using a postMessage event API with auto-height iframe integration, host controls, and lifecycle callbacks for external portals

ACCOMPLISHMENTS

- Kaggle Competitions**
Master Tier Notebook Contributor with 15 Completed Competitions
 - BirdCLEF+ 2026: Ranked 795 out of 4,094 teams by developing an ensemble pipeline combining ProtoSSM and BirdNET models
 - NVIDIA Nemotron Model Reasoning Challenge: Ranked 1,715 out of 4,182 teams in an LLM open model reasoning benchmark
 - Placed in the top 30% in multiple Kaggle competitions, applying gradient boosting, ensemble modeling, feature engineering, and hyperparameter optimization
- August 2025 – Present

- International Quant Championship 2024**
WorldQuant
 - Ranked 165 globally in Stage 1 and 184 globally in Stage 2 among thousands of participants
 - Contributed to a team OS Score of 36,000+ using 24+ financial data fields for quantitative signal generation
 - Collaborated in a two-member team to design and implement alpha strategies under competition constraints
- March 2024 – June 2024

EXTRA-CURRICULAR ACTIVITIES

- Deputy Secretary**
Diplomacia, the IITM Soft Skills Society
- March 2024 – August 2025
SA Contribution Certificate
- Marketing Team Volunteer**
Student Placement Council, IITM BS Degree
- August 2024 – December 2024
SPC Impact Report